

CSCI 198 Project Plan

Title: *Music Intelligence with Deep Learning*

Student: *Noah Wiley*

Department of Computer Science, California State University, Fresno

Supervisor: *David Ruby*

Project Objective:

The objective of this project is to design, implement, and compare multiple deep learning architectures for automated music classification tasks, including genre, mood, and instrumentation recognition. The project aims to evaluate how different model families perform on music audio representations under a unified preprocessing and evaluation framework.

Model Goals:

This project focuses on three primary deep learning model families, each chosen to explore a different aspect of musical structure:

Convolutional Neural Networks (CNNs):

CNNs will serve as the baseline model for this project. They are well-suited for learning patterns from time-frequency representations such as mel-spectrograms. The goal of the CNN models is to establish strong baseline performance on music classification tasks and to evaluate how effectively local spectral features capture musical characteristics

LSTMs:

LSTM-based models will be used to capture temporal dependencies in music. By treating spectrogram frames as sequential inputs, these models aim to learn how musical features evolve over time. The goal is to determine whether explicitly modeling temporal structure improves performance compared to purely convolutional approaches.

Transformer-Based Models:

Transformer models will be implemented to explore long-range temporal relationships using self-attention mechanisms. These models aim to capture global musical structure that may span longer time windows than LSTMs can effectively handle. The goal is to assess whether transformer-based architectures provide performance or interpretability advantages in music classification tasks.

Input Representations:

All models will be trained using standardized audio features extracted from raw audio, including:

- Mel-spectrograms
- Log-mel spectrograms
- MFCCs

Audio will be resampled to a unified sampling rate and segmented into fixed-length clips to ensure consistency across datasets.

Evaluation Criteria:

Models will be evaluated using:

- Accuracy (for single-label tasks)
- Precision, Recall, and F1-score
- Confusion matrices
- Comparative training and validation curves

Performance will be compared across architectures to analyze strengths, weaknesses, and trade-offs.

Expected Outcome:

The expected outcome is a clear comparative analysis of CNN, LSTM, and Transformer models for music classification, supported by quantitative metrics and qualitative error analysis. The results will inform which architectures are best suited for different types of music intelligence tasks.